The development roadmap for the DB R&D project includes creating additional prototypes incorporating the previously mentioned modifications, conducting SEU tests with operational firmware to evaluate the impact of SEU rates on board performance, and analyzing test results to determine the final selection of the SFP+ variant that will be integrated to the final board design. Once the design is adjusted based on the test outcomes, prototypes will be manufactured and tested as part of the production phase. The project aims to produce a total of 930 DB6 units as Stockholm University's contribution to the HL-LHC era for TileCal.

Acknowledgments

This work was supported by Stockholm University and CERN.

References

[1] ATLAS collaboration, *The ATLAS Experiment at the CERN Large Hadron Collider*, 2008 *JINST* **3** S08003.

[2] ATLAS collaboration, *Technical Design Report for the Phase-II Upgrade of the ATLAS Tile Calorimeter*, CERN-LHCC-2017-019, ATLAS-TDR-028 (2018).

[3] E. Valdés Santurio, *Development of the read-out link and control board for the ATLAS Tile Calorimeter Upgrade*, Ph.D. Thesis, Stockholm University (2019) [ISBN: 978-91-7797-896-1] [http://www.diva-portal.org/smash/get/diva2:1364842/FULLTEXT03.pdf].

[4] K. Wyllie, P. Moreira and J. Christiansen, *GBTX MANUAL V0.15 Technical Report*, (2021), https://espace.cern.ch/GBT-Project/GBTX/Manuals/GBTXManual.pdf.

[5] M. Ravnik, *Description of TRIGA Reactor*, https://ric.ijs.si/wp-content/uploads/Description_TRIGA_Reactor.pdf.

[6] Actel, *Radiation-Tolerant ProASIC3 FPGAs Radiation Effects*, (2012), https://www.microsemi.com/document-portal/doc_view/131374-radiation-tolerant-proasic3-fpgas-radiation-effects-report.

[7] D.M. Hiemstra, V. Kirischian and J. Brelski, *Single Event Upset Characterization of the Zynq UltraScale+ MPSoC Using Proton Irradiation*, in *2017 IEEE Radiation Effects Data Workshop (REDW)*, New Orleans, LA, U.S.A. (2017), p. 1–4 [DOI:10.1109/NSREC.2017.8115448].

[8] P. Maillard, J. Barton, M.J. Hart, Y.P. Chen and M.L. Voogel, *Total Ionizing Dose and Single-Events characterization of Xilinx 20nm Kintex UltraScale*, in *2019 19th European Conference on Radiation and Its Effects on Components and Systems (RADECS)*, Montpellier, France (2019), p. 1–6 [DOI:10.1109/RADECS47380.2019.9745695].

[9] ATLAS collaboration, *Radiation Environment of ATLAS*, https://twiki.cern.ch/twiki/pub/AtlasPublic/RadiationSimulationPublicResults/WebRadMaps_Zoom_R2_public.html.

[10] C. Amelung and H. Chen, *ATLAS Radiation Tolerant Electronics Proposed Guideline on COTS Lot to Lot Variation*, (2020), https://indico.cern.ch/event/946481/contributions/3978191/attachments/2087789/3507703/COTSL2L.20200806.v2.pdf.

[11] E. Valdés Santurio et al., *R&D Studies for the Atlas Tile Calorimeter Daughterboard*, *JACoW* **ICALEPCS2021** (2022) 290.